GROUP B9 · FINAL REVIEW · STUDY GUIDE FOR PERSON B

Vehicle Kinematics: Every Formula, Derived

Part 2 of the presentation: how the rig moves for a given speed and steering angle, where each equation comes from, and why reversing is unstable. Every step of the maths is written out, so you can reproduce it on a board and explain what each term means.

Your slides: 9–13 Your time: about 5 minutes From Person A (model) · to Person C (algorithms)

1. The whole project in one page
2. What your part must achieve
3. The maths toolkit
4. The tractor: kinematic bicycle model
5. The trailer: hitch and no-slip
6. The hitch-angle equation
7. The complete model, and how we simulate it
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1 The Whole Project in One Page

Read this first. Every presenter should be able to tell this story in one minute, because the professor may ask any of you how your part fits in.

The Problem

Cars such as the Mahindra BE 6 and XEV 9e can already park themselves (Autopark: 12 ultrasonic sensors, perpendicular, angled and parallel bays). A truck with a trailer cannot, and it is the vehicle that needs help most. The reason is physics: a tractor-trailer is two bodies joined at a hitch, and **when it reverses, the hitch angle grows on its own**. A 2° error becomes 60° after 28 m of reversing. If nobody corrects it, the trailer folds against the cab: a **jackknife**.

What We Built

A guidance system that takes a tractor-trailer from the yard gate into a chosen bay while the hitch angle never exceeds 60°. It has four layers:

Layer	What it does	Presented in
Vehicle model	Four numbers describe the rig; four equations say how they change for a given speed and steering	Part 1 (A), Part 2 (B)
Hybrid A* planner	Searches the yard once and returns a route the truck can actually drive, gate to bay	Part 3 (C)
NMPC controller	Every 0.25 s, predicts 9.6 s ahead and chooses the steering that follows the route with |ψ| ≤ 60° as a hard limit	Part 3 (C)
Driver guidance	Turns the controller's output into gear, speed (km/h), steering-wheel turns and distance	Part 4 (D)

The Result

We tested 110 free bays (4 layouts × 6 random fills of parked trucks). All 110 were parked within tolerance, the hitch limit was never broken (worst case 50.1°), and a driver model without hitch awareness jackknifed on all 72 reversing manoeuvres.

The Chain of Ideas

THE STORY IN FOUR SENTENCES

(1) We model the truck as a wheeled robot with four state variables and two inputs. (2) Its kinematics show that reversing is unstable: the hitch angle grows exponentially. (3) So we plan a drivable route with Hybrid A* and follow it with an NMPC that looks ahead and treats the hitch limit as a hard constraint. (4) The result parks every test bay and tells a human driver exactly what to do.

Symbol	Meaning	Value
x_1, y_1	Position of the tractor's rear-axle centre	metres
$\theta_0,\ \theta_1$	Heading of the tractor, heading of the trailer (from the x axis, anticlockwise positive)	radians or degrees
$\psi = \theta_0 - \theta_1$	Hitch angle: how far the tractor is turned relative to the trailer	hard limit 60°
$v,\ \delta$	Speed (negative in reverse), front-wheel steering angle	$\lvert v\rvert \le 2$ m/s, $\lvert\delta\rvert \le 35^\circ$
$L_1,\ L_2,\ d$	Tractor wheelbase, hitch to trailer axle, hitch ahead of the rear axle	4 m, 8 m, 0.5 m

2 What Your Part Must Achieve

Person A gives the professor the symbols: state $X = [x_1, y_1, \theta_0, \psi]^T$, input $U = [v, \delta]^T$. You give him the **four equations** $\dot{X} = f(X, U)$, show where each comes from, and end with the single most important fact of the project:

YOUR HEADLINE

Linearising the hitch equation gives $\dot{\psi} \approx -\frac{v}{L_2}\psi$. The eigenvalue $\lambda = -v/L_2$ is negative going forward (stable) and **positive in reverse (unstable)**: a 2° error grows to 60° in 28 m. That is why Part 3 needs a planner and a predictive controller.

The four equations you will derive:

$$\begin{aligned}
\dot{x}_1 &= v\cos\theta_0 \\
\dot{y}_1 &= v\sin\theta_0 \\
\dot{\theta}_0 &= \frac{v}{L_1}\tan\delta \\
\dot{\psi} &= v\left[\frac{\tan\delta}{L_1} - \frac{\sin\psi}{L_2} - \frac{d\,\tan\delta\,\cos\psi}{L_1 L_2}\right]
\end{aligned}$$

3 The Maths Toolkit

Everything in this guide uses only the tools below. Learn them first; the derivations are then just bookkeeping.

3.1 Heading and Sideways Unit Vectors

$$\hat{u}(\theta) = \begin{pmatrix}\cos\theta \\ \sin\theta\end{pmatrix}, \qquad \hat{n}(\theta) = \begin{pmatrix}-\sin\theta \\ \cos\theta\end{pmatrix}$$

Both have length 1 and are perpendicular: $\hat{u}\cdot\hat{n} = 0$.

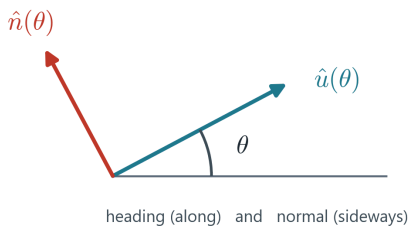


Figure 1. $\hat{u}(\theta)$ points along a heading, $\hat{n}(\theta)$ is $\hat{u}$ turned 90° anticlockwise.

3.2 Differentiating a Rotating Unit Vector

If the heading changes with time, $\theta = \theta(t)$, the chain rule gives

$$\frac{d}{dt}\hat{u}(\theta) = \begin{pmatrix}-\sin\theta \\ \cos\theta\end{pmatrix}\dot{\theta} = \dot{\theta}\,\hat{n}(\theta)$$

Meaning: when a body rotates at rate $\dot{\theta}$, a point a distance r along its axis moves sideways at speed $r\dot{\theta}$. This one rule produces every rotation term in this guide.

3.3 Dot Products Between Two Headings

The dot product picks out the component of one vector along another. Using $\cos(a-b) = \cos a\cos b + \sin a\sin b$ and $\sin(a-b) = \sin a\cos b - \cos a\sin b$:

$$\hat{u}(a)\cdot\hat{u}(b)=\cos a\cos b+\sin a\sin b=\cos(a-b)$$
$$\hat{u}(a)\cdot\hat{n}(b)=-\cos a\sin b+\sin a\cos b=\sin(a-b)$$
$$\hat{n}(a)\cdot\hat{n}(b)=\sin a\sin b+\cos a\cos b=\cos(a-b)$$

With $a=\theta_0$ and $b=\theta_1$, the difference $a-b$ is exactly the hitch angle ψ. That is why $\sin\psi$ and $\cos\psi$ appear in the trailer equation.

3.4 Small-Angle Approximations

For an angle x in **radians** close to 0: $\sin x\approx x$, $\cos x\approx 1$, $\tan x\approx x$. (At 10° the error of $\sin x\approx x$ is only 0.5 %.) Always use radians in formulas: $1^\circ=\pi/180\approx 0.01745$ rad.

3.5 The Exponential Solution

The equation $\dot{\psi}=\lambda\psi$ has the solution $\psi(t)=\psi_0\,e^{\lambda t}$. If $\lambda<0$, ψ dies out; if $\lambda>0$, it grows, multiplying by $e\approx 2.72$ every $1/\lambda$ seconds and doubling every $\ln 2/\lambda$ seconds.

4 The Tractor: Kinematic Bicycle Model

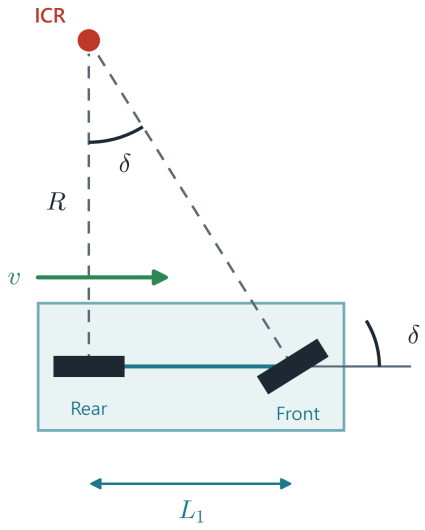


Figure 2. Each axle's two wheels are lumped into one wheel on the centre line. Both wheel axles meet at the instantaneous centre of rotation (ICR).

4.1 The Bicycle Simplification

A real tractor has two front wheels and two (or four) rear wheels. If the steering is designed properly (Ackermann geometry), all wheels turn about one common point, and the left and right wheels of an axle behave like a single wheel at the axle's centre. So we draw the tractor as a bicycle: one rear wheel, one steered front wheel, L_1 apart.

4.2 Rear Axle: the First Two Equations

The rear wheel cannot slide sideways (Person A's no-slip constraint), so its velocity points exactly along the heading:

$$\dot{p}_1 = v\,\hat{u}(\theta_0) \quad\Longrightarrow\quad \dot{x}_1 = v\cos\theta_0, \qquad \dot{y}_1 = v\sin\theta_0$$

This is also the definition of v: the signed speed of the rear-axle centre, negative in reverse.

4.3 Front Axle: the Turning-Rate Equation, by Vectors

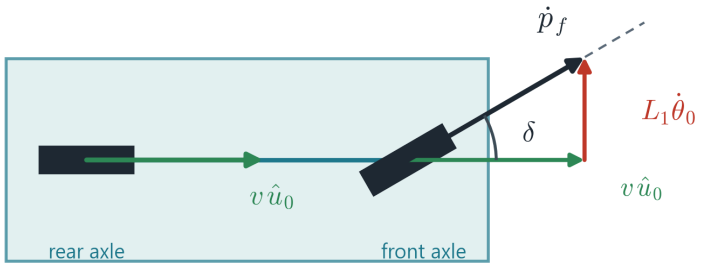


Figure 3. The front axle moves with the rear axle ($v\hat{u}_0$) plus a sideways part from the rotation ($L_1\dot{\theta}_0\hat{n}_0$). The sum must point along the front wheel.

1. Front-axle position: $p_f = p_1 + L_1\,\hat{u}(\theta_0)$.
2. Differentiate, using rule 3.2:
$$\dot{p}_f = v\,\hat{u}(\theta_0) + L_1\dot{\theta}_0\,\hat{n}(\theta_0)$$
3. The front wheel points at $\theta_0 + \delta$ and cannot slide sideways, so the velocity has no component along $\hat{n}(\theta_0 + \delta)$:
$$\dot{p}_f \cdot \hat{n}(\theta_0 + \delta) = 0$$
4. Use the dot products of 3.3 with $a = \theta_0$, $b = \theta_0 + \delta$, so $a - b = -\delta$:
$$\hat{u}(\theta_0)\cdot\hat{n}(\theta_0+\delta) = \sin(-\delta) = -\sin\delta, \qquad \hat{n}(\theta_0)\cdot\hat{n}(\theta_0+\delta) = \cos\delta$$
5. Substitute:
$$-v\sin\delta + L_1\dot{\theta}_0\cos\delta = 0 \quad\Longrightarrow\quad \boxed{\dot{\theta}_0 = \frac{v}{L_1}\tan\delta}$$

4.4 The Same Result from the ICR Geometry

Every wheel rolls on a circle around the ICR, so each wheel's axle line passes through it. The rear axle line is perpendicular to the body; the front axle line is perpendicular to the front wheel. In Figure 2 this makes a right-angled triangle with the body L_1 as one side and the turning radius R (rear axle to ICR) as the other, and the angle at the ICR equals δ (two lines each turned by δ make the same angle). So

$$\tan\delta = \frac{L_1}{R} \;\Longrightarrow\; R = \frac{L_1}{\tan\delta}, \qquad \dot{\theta}_0 = \frac{v}{R} = \frac{v}{L_1}\tan\delta$$

The second step is the rule for any circle: angular rate = speed ÷ radius. Both methods agree; use whichever you can draw faster on a board.

Numbers: at full lock, $\delta = 35^\circ$: $R = 4/\tan 35^\circ = 4/0.700 = 5.71$ m, the tightest circle the rear axle can drive. At $\delta = 0$: $R \to \infty$, straight line, $\dot\theta_0 = 0$.

5 The Trailer: Hitch and No-Slip

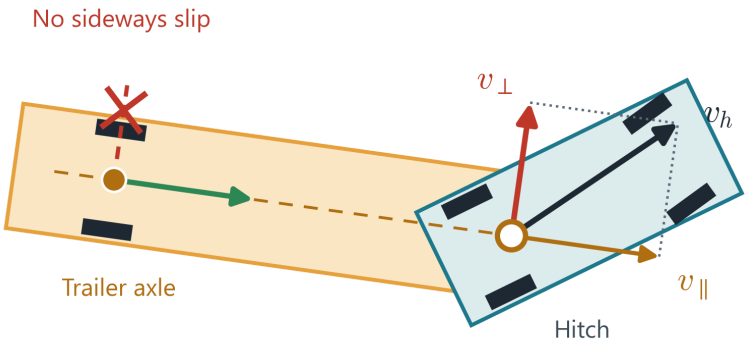


Figure 4. The hitch velocity v_h split into a part along the trailer ($v_\parallel$) and a part across it ($v_\perp$). The trailer axle cannot slide, so only $v_\perp$ can rotate the trailer, about its own axle.

1. **Hitch position and velocity.** The hitch is d ahead of the rear axle on the tractor:

$$p_h = p_1 + d\,\hat u(\theta_0) \quad\Longrightarrow\quad \dot p_h = v\,\hat u(\theta_0) + d\,\dot\theta_0\,\hat n(\theta_0)$$

2. **Trailer-axle position and velocity.** The trailer axle is L_2 behind the hitch along the trailer:

$$p_2 = p_h - L_2\,\hat u(\theta_1) \quad\Longrightarrow\quad \dot p_2 = \dot p_h - L_2\,\dot\theta_1\,\hat n(\theta_1)$$

3. **No-slip at the trailer axle:** no velocity along the trailer's sideways direction:

$$\dot p_2 \cdot \hat n(\theta_1) = 0 \implies \dot p_h \cdot \hat n(\theta_1) - L_2\,\dot\theta_1 \underbrace{\hat n(\theta_1)\cdot\hat n(\theta_1)}_{=1} = 0 \implies \dot\theta_1 = \frac{\dot p_h \cdot \hat n(\theta_1)}{L_2}$$

   The numerator is exactly $v_\perp$ in Figure 4: the sideways part of the hitch velocity.

4. **Compute** $v_\perp$ with 3.3, $a = \theta_0$, $b = \theta_1$, $a - b = \psi$:

$$v_\perp = v\underbrace{\hat u(\theta_0)\cdot\hat n(\theta_1)}_{\sin\psi} + d\,\dot\theta_0\underbrace{\hat n(\theta_0)\cdot\hat n(\theta_1)}_{\cos\psi} = v\sin\psi + d\,\dot\theta_0\cos\psi$$

5. **Result:**

$$\boxed{\dot\theta_1 = \frac{v\sin\psi + d\,\dot\theta_0\cos\psi}{L_2}}$$

SAY IT LIKE THIS

"The trailer is dragged by its hitch. Split the hitch velocity into a part along the trailer and a part across it. The trailer axle can only roll, so the along part just pulls the trailer straight, and the across part swings it about its axle, with lever arm L_2. That gives $\dot{\theta}_1 = v_\perp / L_2$."

Extra, if asked: the trailer axle's own speed is the along part, $v_2 = \dot{p}_h \cdot \hat{u}(\theta_1) = v\cos\psi - d\,\dot{\theta}_0 \sin\psi$. For a straight rig ($\psi = 0$) it equals v, as expected.

6 The Hitch-Angle Equation

6.1 Assemble It

Since $\psi = \theta_0 - \theta_1$, its rate is $\dot{\psi} = \dot{\theta}_0 - \dot{\theta}_1$. Substitute $\dot{\theta}_0 = \frac{v}{L_1}\tan\delta$ into both places:

$$\dot{\psi} = \frac{v}{L_1}\tan\delta - \frac{v\sin\psi}{L_2} - \frac{d\cos\psi}{L_2}\cdot\frac{v}{L_1}\tan\delta = v\left[\frac{\tan\delta}{L_1} - \frac{\sin\psi}{L_2} - \frac{d\,\tan\delta\,\cos\psi}{L_1 L_2}\right]$$

A useful factored form (group the two steering terms):

$$\dot{\psi} = v\left[\frac{\tan\delta}{L_1}\left(1 - \frac{d}{L_2}\cos\psi\right) - \frac{\sin\psi}{L_2}\right]$$

6.2 What Each Term Means

Term	Physical meaning	Size in our rig
$\frac{\tan\delta}{L_1}$ (steering)	The tractor's own turning. It is the **only term the driver controls**.	Up to $\tan 35^\circ/4 = 0.175$ per m
$-\frac{\sin\psi}{L_2}$ (hitch)	The trailer's reaction: it swings to line up behind the hitch. Restoring forward, **destabilising in reverse** (Section 8).	Up to $1/8 = 0.125$ per m
$-\frac{d\tan\delta\cos\psi}{L_1 L_2}$ (offset)	Appears only because the hitch is ahead of the rear axle; when the tractor turns, the hitch moves sideways and nudges the trailer.	$d/L_2 = 6\%$ of the steering term

6.3 Sanity Checks (Do These Out Loud If Asked)

Check	What the formula gives	Makes sense?
Units	v [m/s] × [1/m] = [1/s]: a rate of angle in rad/s	Yes
Parked, $v = 0$	$\dot{\psi} = 0$: nothing moves	Yes
Straight rig, straight wheels ($\psi = 0$, $\delta = 0$)	$\dot{\psi} = 0$: it stays straight	Yes
Hitch over the axle, $d = 0$	$\dot{\psi} = v\left[\frac{\tan\delta}{L_1} - \frac{\sin\psi}{L_2}\right]$: the textbook form	Yes
Very long trailer, $L_2 \to \infty$	$\dot{\psi} \to \dot{\theta}_0$: the trailer barely turns	Yes
Reverse gear, $v \to -v$	Every term flips sign	See Section 8

7 The Complete Model, and How We Simulate It

7.1 Every Term Is Proportional to v

Look at the four equations: each right-hand side has a factor v. Divide by v and write $ds = v\,dt$ for the (signed) distance travelled:

$$\frac{dx_1}{ds} = \cos\theta_0, \quad \frac{dy_1}{ds} = \sin\theta_0, \quad \frac{d\theta_0}{ds} = \frac{\tan\delta}{L_1}, \quad \frac{d\psi}{ds} = \frac{\tan\delta}{L_1}\left(1 - \frac{d}{L_2}\cos\psi\right) - \frac{\sin\psi}{L_2}$$

Consequence: the *shape* of the path depends only on the steering along the distance, not on how fast you drive. Driving the same steering twice as fast traces the same path in half the time. That is why the planner chooses the path first and a speed profile separately.

7.2 Integrating with RK4

The equations give rates; to predict the future we integrate them in small time steps Δt. We use the classical fourth-order Runge-Kutta method (RK4), holding v and δ constant over each step:

$$\begin{aligned} k_1 &= f(X_k), & k_2 &= f\left(X_k + \tfrac{\Delta t}{2} k_1\right), \\ k_3 &= f\left(X_k + \tfrac{\Delta t}{2} k_2\right), & k_4 &= f(X_k + \Delta t\, k_3), \\ X_{k+1} &= X_k + \tfrac{\Delta t}{6}\left(k_1 + 2k_2 + 2k_3 + k_4\right) \end{aligned}$$

It samples the slope four times per step and averages them, so its error per step shrinks like Δt^5 (Euler's method, which uses only k_1, shrinks like Δt^2). We use 0.05 s steps in the simulator, 0.4 s steps inside the NMPC prediction and 0.5 m steps in the planner's motion primitives. One model, one integrator, shared by planner, controller and viewer.

8 Why Reversing Is Unstable

8.1 Linearise Around the Straight Rig

Hold the steering straight, $\delta = 0$. The hitch equation becomes

$$\dot{\psi} = -\frac{v}{L_2}\sin\psi$$

Its equilibrium is $\psi = 0$. Near it, $\sin\psi \approx \psi$:

$$\dot{\psi} \approx \lambda\,\psi, \qquad \lambda = -\frac{v}{L_2}, \qquad \psi(t) = \psi_0\, e^{\lambda t}$$

Gear	Eigenvalue	Behaviour (at 1 m/s, L_2 = 8 m)
Forward, $v > 0$	$\lambda = -v/L_2 < 0$	**Stable.** Any hitch angle decays with time constant L_2/v = 8 s. A towed trailer straightens itself.
Reverse, $v < 0$	$\lambda = \lvert v\rvert/L_2 > 0$	**Unstable.** The angle grows ×e every 8 m, doubles every $L_2 \ln 2 = 5.5$ m.

With steering included, the linearisation is $\dot{\psi} \approx -\frac{v}{L_2}\psi + \frac{v}{L_1}\left(1 - \frac{d}{L_2}\right)\delta$: the first term is the instability, the second is how steering pushes on it.

SAY IT LIKE THIS

"The hitch term is minus v over L_2 times ψ. Going forward it pulls ψ back to zero, like a trailer following the truck. In reverse v is negative, the sign flips, and the term pushes ψ away from zero, like balancing a broom on your hand. The eigenvalue is positive, so any small error grows exponentially."

8.2 The Exact Solution (Not Only Near Zero)

We do not need the approximation. Write the equation in distance: reversing, $ds = |v|\,dt$ and $v = -|v|$, so

$$\frac{d\psi}{ds} = \frac{\sin\psi}{L_2} \quad\Longrightarrow\quad \int \frac{d\psi}{\sin\psi} = \int \frac{ds}{L_2}$$

The standard integral is $\int \frac{d\psi}{\sin\psi} = \ln\left|\tan\frac{\psi}{2}\right|$. You can check it by differentiating, using $\sin\psi = 2\sin\frac{\psi}{2}\cos\frac{\psi}{2}$:

$$\frac{d}{d\psi}\ln\tan\frac{\psi}{2} = \frac{1}{\tan\frac{\psi}{2}}\cdot\frac{1}{2\cos^2\frac{\psi}{2}} = \frac{1}{2\sin\frac{\psi}{2}\cos\frac{\psi}{2}} = \frac{1}{\sin\psi}$$

So

$$\ln\tan\frac{\psi}{2} - \ln\tan\frac{\psi_0}{2} = \frac{s}{L_2} \quad\Longrightarrow\quad \boxed{\tan\frac{\psi}{2} = \tan\frac{\psi_0}{2}\; e^{\,s/L_2}} \qquad \text{(reverse)}$$

Forward, the exponent is $-s/L_2$ and the angle decays. For small angles, $\tan(\psi/2) \approx \psi/2$ and this becomes $\psi \approx \psi_0 e^{s/L_2}$, the linear result again.

8.3 The Numbers on the Slide

Solve for the distance: $s = L_2 \ln \dfrac{\tan(\psi/2)}{\tan(\psi_0/2)}$. Starting from a 2° error ($\psi_0/2 = 1^\circ$):

Reach	Calculation	Distance reversed
60° (our hard limit)	$8 \ln \dfrac{\tan 30^\circ}{\tan 1^\circ} = 8 \ln \dfrac{0.5774}{0.01746} = 8 \ln 33.08 = 8 \times 3.499$	**28.0 m**
75° (physical jackknife)	$8 \ln \dfrac{\tan 37.5^\circ}{\tan 1^\circ} = 8 \ln 43.96 = 8 \times 3.783$	**30.3 m**

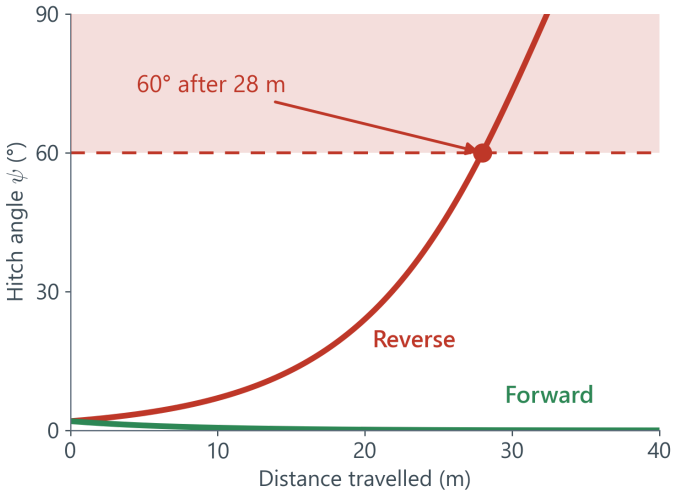


Figure 5. The exact solution from 2°: reversing passes 60° after 28 m; forward, the angle vanishes. Our simulator gives the same numbers, which is a good check that the code matches the maths.

Why this matters: a bay approach is 20 to 40 m of reversing. Without correction, any small error becomes a jackknife within one manoeuvre. A controller must act early, before the error grows, which means it must *look ahead*: that is the motivation for NMPC in Part 3.

9 How Much Steering Holds a Hitch Angle

To hold the hitch at a constant angle (for example while reversing round a curve), set $\dot{\psi} = 0$ in the factored equation (with $v \neq 0$):

$$\frac{\tan\delta}{L_1}\left(1 - \frac{d}{L_2}\cos\psi\right) = \frac{\sin\psi}{L_2} \quad \Longrightarrow \quad \tan\delta_{\text{hold}} = \frac{L_1}{L_2} \cdot \frac{\sin\psi}{1 - \frac{d}{L_2}\cos\psi}$$

At ψ = 60°: $\tan\delta = 0.5 \times \dfrac{0.866}{1 - 0.0625 \times 0.5} = \dfrac{0.433}{0.969} = 0.447$, so $\delta_{\text{hold}} = 24.1^\circ$.

The largest angle steering can hold. Let $c = d/L_2 = 0.0625$ and $g(\psi) = \dfrac{\sin\psi}{1 - c\cos\psi}$. Differentiate (quotient rule):

$$g'(\psi) = \frac{\cos\psi\,(1 - c\cos\psi) - \sin\psi\,(c\sin\psi)}{(1 - c\cos\psi)^2} = \frac{\cos\psi - c}{(1 - c\cos\psi)^2}$$

(using $\cos^2 + \sin^2 = 1$). It is zero at $\cos\psi^* = c$, so $\psi^* = 86.4^\circ$, where $g = 1/\sqrt{1 - c^2} = 1.002$. Then $\tan\delta_{\max} = 0.5 \times 1.002$, so $\delta = 26.6^\circ$.

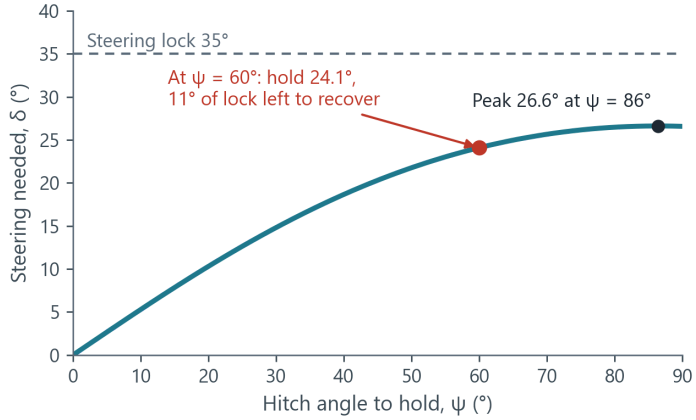


Figure 6. Steering needed to hold each hitch angle. Holding 60° already costs 24.1° of our 35° lock.

WHY THE LIMIT IS 60°

To *straighten* a hitch you must steer more than the holding angle. At 60° holding needs 24.1°, leaving about 11° of lock to pull the trailer back. Much beyond 60° there is too little margin, and past about 86° no steering can even hold the angle. So the controller treats 60° as a hard limit and the planner stays below 52°.

10 Why Reversing Feels Backwards

Drivers learn "to reverse the trailer left, first steer right". The equations show why. Reverse ($v = -1$ m/s) with constant left steering $\delta = +10^\circ$ from a straight rig:

- $\dot\theta_0 = v\tan\delta/L_1 < 0$: the **tractor** heading turns clockwise straight away.
- That makes ψ negative, and then $\dot\theta_1 \approx v\sin\psi/L_2 > 0$: the **trailer** heading turns the *other* way.

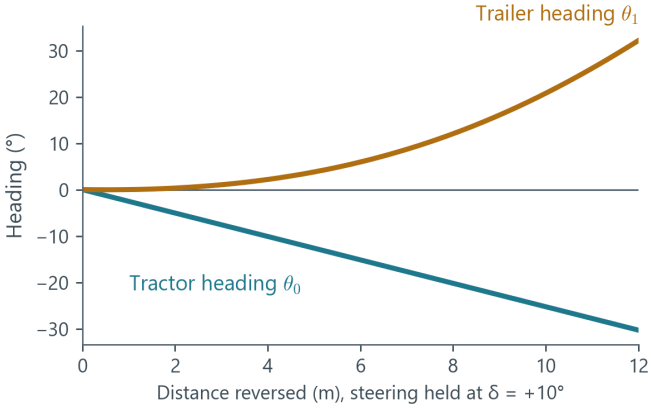


Figure 7. Reversing 12 m with the wheel held at +10° (our model): the tractor ends at −30°, the trailer at +32°.

In control language, the tractor's rear axle is a **non-minimum-phase** output in reverse: to move the trailer one way, the tractor must first move the other way. A controller that only reacts to the current error (like PID) steers the wrong way at first. This is why our NMPC tracks the **trailer axle** (Person C will mention it).

11 Worked Example: Evaluate the Model at One Instant

State: $\theta_0 = 10^\circ$, $\psi = 20^\circ$. Input: reversing at $v = -1$ m/s with $\delta = +15^\circ$. ($\tan 15^\circ = 0.2679$, $\sin 20^\circ = 0.3420$, $\cos 20^\circ = 0.9397$.)

Quantity	Calculation	Result
$\dot{x}_1$	$-1 \times \cos 10^\circ = -0.985$	−0.985 m/s
$\dot{y}_1$	$-1 \times \sin 10^\circ = -0.174$	−0.174 m/s
$\dot{\theta}_0$	$-1 \times 0.2679/4$	−0.0670 rad/s (−3.84°/s)
Steering term	$0.2679/4$	0.06699
Hitch term	$0.3420/8$	0.04275
Offset term	$0.5 \times 0.2679 \times 0.9397/32$	0.00393
$\dot{\psi}$	$-1 \times (0.06699 - 0.04275 - 0.00393)$	−0.0203 rad/s (−1.16°/s)
$\dot{\theta}_1$ (check)	$(-1 \times 0.3420 + 0.5 \times (-0.0670) \times 0.9397)/8$	−0.0467 rad/s; $\dot{\theta}_0 - \dot{\theta}_1 = -0.0203$ ✓

Reading it: with the hitch at +20°, reversing with 15° of left steering *reduces* the hitch angle by about 1.2° per second: in reverse, steering towards the side the tractor is turned straightens the rig. With straight wheels the same state would give $\dot{\psi} = +0.0428$ rad/s: the angle would grow.

12 The Derivation on a Board, in Six Lines

1. $\hat{u} = (\cos\theta, \sin\theta)$, $\hat{n} = (-\sin\theta, \cos\theta)$, $\frac{d}{dt}\hat{u} = \dot{\theta}\,\hat{n}$.
2. Rear axle rolls: $\dot{p}_1 = v\,\hat{u}_0 \Rightarrow \dot{x}_1 = v\cos\theta_0$, $\dot{y}_1 = v\sin\theta_0$.
3. Front wheel rolls: $(v\hat{u}_0 + L_1\dot{\theta}_0\hat{n}_0)\cdot\hat{n}(\theta_0+\delta) = 0 \Rightarrow -v\sin\delta + L_1\dot{\theta}_0\cos\delta = 0 \Rightarrow \dot{\theta}_0 = v\tan\delta/L_1$.
4. Hitch: $\dot{p}_h = v\hat{u}_0 + d\dot{\theta}_0\hat{n}_0$. Trailer axle rolls: $\dot{p}_2\cdot\hat{n}_1 = 0 \Rightarrow \dot{\theta}_1 = \dot{p}_h\cdot\hat{n}_1/L_2$.
5. $\hat{u}_0\cdot\hat{n}_1 = \sin\psi$, $\hat{n}_0\cdot\hat{n}_1 = \cos\psi \Rightarrow \dot{\theta}_1 = (v\sin\psi + d\dot{\theta}_0\cos\psi)/L_2$.
6. $\dot{\psi} = \dot{\theta}_0 - \dot{\theta}_1 = v\left[\frac{\tan\delta}{L_1} - \frac{\sin\psi}{L_2} - \frac{d\tan\delta\cos\psi}{L_1 L_2}\right]$; at $\delta = 0$, small ψ: $\dot{\psi} \approx -\frac{v}{L_2}\psi$.

13 What to Say on Each Slide

Slide 9 · Part 2 divider (10 s)
"Person A gave us four state variables and two inputs. I will show how those four numbers change, and why reversing is unstable."

Slide 10 · The Tractor: Kinematic Bicycle Model (about 1 min)
Lump each axle into one wheel. The wheel axles meet at the ICR; $R = L_1/\tan\delta$. Rear axle moves along its heading, so $\dot{x}_1 = v\cos\theta_0$, $\dot{y}_1 = v\sin\theta_0$; it moves on a circle of radius R, so $\dot{\theta}_0 = v/R = v\tan\delta/L_1$.

Slide 11 · The Trailer: Hitch Geometry and No-Slip Constraint (about 1.5 min)
Hitch position, trailer-axle position (rigid-body transforms). Split the hitch velocity; only $v_\perp$ turns the trailer about its axle: $\dot{\theta}_1 = v_\perp/L_2$ with $v_\perp = v\sin\psi + d\dot{\theta}_0\cos\psi$.

Slide 12 · The Complete Four-State Model (about 1 min)
$\dot{\psi} = \dot{\theta}_0 - \dot{\theta}_1$; read the three colour-coded terms. Every term has a factor v: path shape depends on distance, not speed. RK4, one model shared by planner, controller and viewer.

Slide 13 · Why Reversing Is Unstable (about 1.5 min)
Linearise: $\lambda = -v/L_2$. Forward stable, reverse unstable. Exact solution: 2° to 60° in 28 m. "So a controller must look ahead and correct early. Person C will show how."

14 Questions the Professor May Ask

Derive $\dot{\psi}$ on the board.
Use the six lines in Section 12. Start from the no-slip constraints, not from the final formula.

Why does $\tan\delta$ appear and not δ?
From the ICR triangle: $\tan\delta = L_1/R$. For small angles $\tan\delta \approx \delta$, but at 35° the difference is 14 %.

What is the eigenvalue and what does it mean?
$\lambda = -v/L_2$ from linearising at $\psi = 0$, $\delta = 0$. Its sign decides stability: negative forward, positive in reverse. $1/|\lambda|$ is the time for a ×e change: 8 s at 1 m/s.

Is the system stable forward?
The hitch angle is: a trailer straightens behind a truck driving forward. Position and heading are not "stable" in any sense; they just follow the steering.

Why does speed not appear in the 28 m figure?
Every term is proportional to v, so in terms of distance the growth is the same at any speed: ×e every L_2 metres. Faster reversing only makes it happen sooner in time.

What does the offset d do?
With the hitch ahead of the rear axle, steering moves the hitch sideways, which slightly rotates the trailer the same way. It reduces the steering term by $(1 - \frac{d}{L_2}\cos\psi)$, about 6 %.

Why 60° as the limit?
Holding 60° needs 24.1° of steering, leaving ~11° of our 35° lock to recover; beyond about 86° no steering can hold the angle. The physical jackknife is 75°.

What is non-minimum phase here?
In reverse, the tractor initially moves opposite to where the trailer must go (Figure 7). Reactive control steers the wrong way first; we track the trailer axle instead.

Why RK4 and not Euler?

Fourth-order accuracy at the 0.4 s prediction steps the NMPC uses; Euler would need much smaller steps for the same accuracy. RK4 costs four evaluations of a cheap function.

What if the wheels slip?

Then the no-slip constraints, and the model, break. At ≤ 2 m/s on dry asphalt slip is centimetres, inside our 0.25 m clearance. Ice or speed would need a dynamic tyre model.

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15 Check Yourself

1. Compute the turning radius of the rear axle at $\delta = 20^\circ$.
2. Show that $\hat{n}(a) \cdot \hat{u}(b) = -\sin(a - b)$.
3. With $\delta = 0$ and $\psi = 10^\circ$, reversing at 1 m/s, what is $\dot{\psi}$ in °/s? Is the angle growing or shrinking?
4. How many metres of reversing does it take for a 1° hitch error to reach 60° (straight wheels)?
5. What steering holds $\psi = 30^\circ$?
6. If L_2 were 12 m instead of 8 m, would reversing be more or less forgiving? By how much?

Answers

1. $R = 4/\tan 20^\circ = 4/0.364 = 11.0$ m.
2. $\hat{n}(a) \cdot \hat{u}(b) = -\sin a \cos b + \cos a \sin b = -(\sin a \cos b - \cos a \sin b) = -\sin(a - b)$.
3. $\dot{\psi} = -\frac{v}{L_2} \sin\psi = -\frac{-1}{8} \times 0.1736 = +0.0217$ rad/s $= +1.24$°/s. Positive and ψ is positive, so it is growing.
4. $s = 8 \ln \frac{\tan 30^\circ}{\tan 0.5^\circ} = 8 \ln \frac{0.5774}{0.008727} = 8 \ln 66.2 = 33.5$ m.
5. $\tan\delta = 0.5 \times \frac{0.5}{1 - 0.0625 \times 0.866} = \frac{0.25}{0.946} = 0.264$, so $\delta = 14.8^\circ$.
6. More forgiving: the error grows ×e every L_2 metres, so every 12 m instead of 8 m (50 % more distance). Longer trailers are easier to reverse, which drivers also say.

16 One-Page Cheat Sheet

The Model

$$\dot{x}_1 = v\cos\theta_0, \quad \dot{y}_1 = v\sin\theta_0, \quad \dot{\theta}_0 = \frac{v}{L_1}\tan\delta, \quad \dot{\psi} = v\left[\frac{\tan\delta}{L_1} - \frac{\sin\psi}{L_2} - \frac{d\tan\delta\cos\psi}{L_1 L_2}\right]$$

Where Each Comes From

Rear axle rolls along its heading. Front wheel rolls along $\theta_0 + \delta \Rightarrow R = L_1/\tan\delta$, $\dot{\theta}_0 = v/R$. Trailer axle rolls $\Rightarrow \dot{\theta}_1 = v_\perp/L_2$, $v_\perp = v\sin\psi + d\dot{\theta}_0\cos\psi$. Then $\dot{\psi} = \dot{\theta}_0 - \dot{\theta}_1$.

Tools

$\frac{d}{dt}\hat{u} = \dot{\theta}\hat{n}$; $\hat{u}(a)\cdot\hat{n}(b) = \sin(a-b)$; $\hat{n}(a)\cdot\hat{n}(b) = \cos(a-b)$.

Instability

$\delta = 0$, small ψ: $\dot{\psi} \approx -\frac{v}{L_2}\psi$, $\lambda = -v/L_2$. Forward stable, reverse unstable: ×e every 8 m, doubles every 5.5 m. Exact: $\tan\frac{\psi}{2} = \tan\frac{\psi_0}{2}e^{s/L_2}$; 2° → 60° in 28.0 m, → 75° in 30.3 m.

Holding and the Limit

$\tan\delta_{\text{hold}} = \frac{L_1}{L_2}\frac{\sin\psi}{1-\frac{d}{L_2}\cos\psi}$: 24.1° at 60°, maximum 26.6° (at 86°). Limit 60° leaves ~11° of 35° lock to recover.

Numbers

L_1 = 4 m, L_2 = 8 m, d = 0.5 m, $R_{\min}$ = 5.71 m at 35°. RK4: 0.05 s (simulation), 0.4 s (NMPC), 0.5 m (planner).